



UNIVERSITY OF
WATERLOO

MTE 203 – Advanced Calculus

Midterm Exam Practice Problem 1

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The surfaces

$$\begin{aligned}x^2 + y^2 - 2 &= 0 \\x + z - 4 &= 0\end{aligned}$$

meet in an ellipse E.

Find the equation of the line tangent to E at $P_0(1,1,3)$

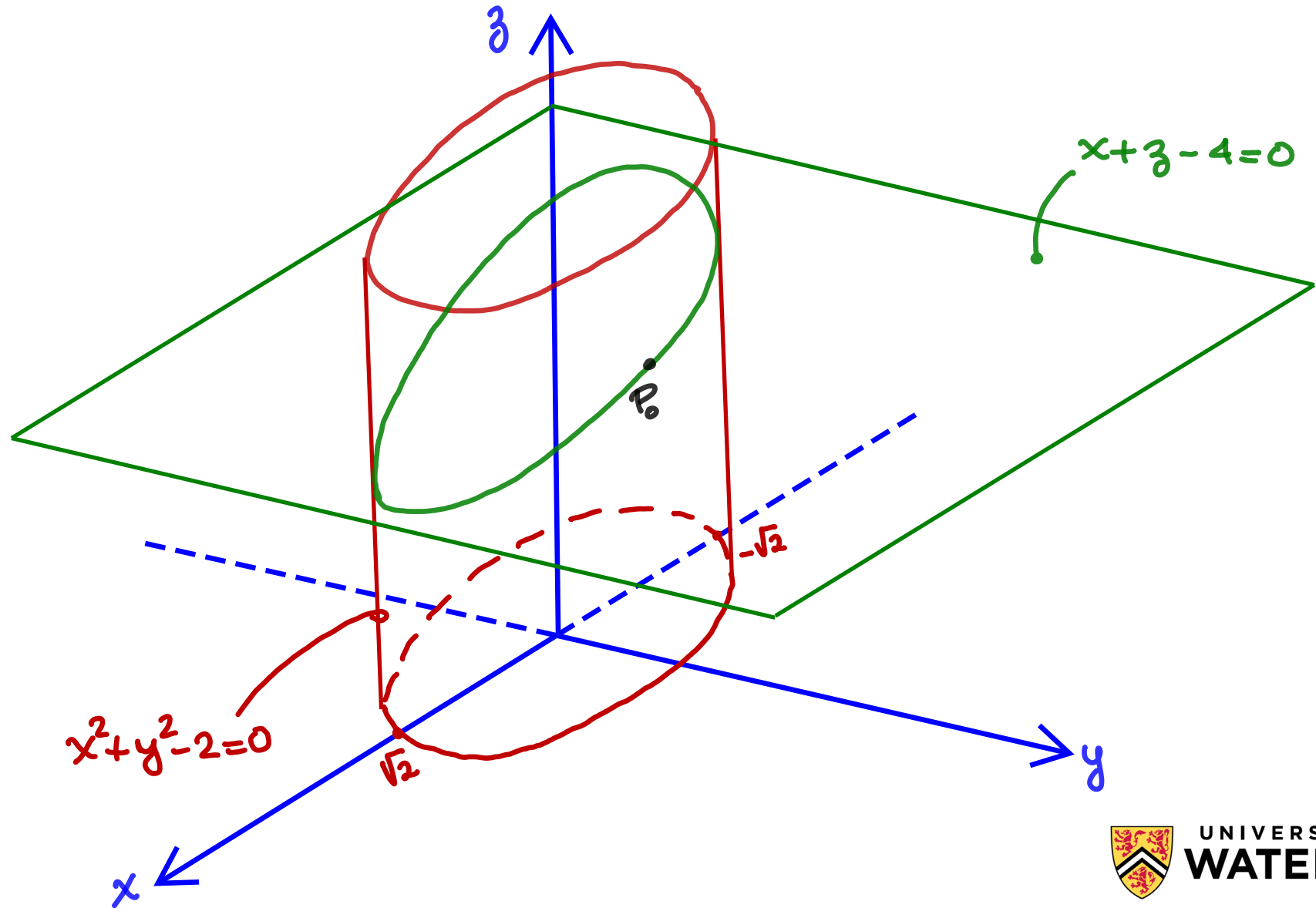
Solution:

Index: Gradient and Tangent Line



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